NOLTA 投稿用実験（その 4）

k は固定したもとで，次元数 d を変化させてみる．

実験の条件は各種変数がそれぞれ小さい値として行う．（余裕があれば増やしても良いかも．）

繰り返して平均を取る必要あり．

1. パラメータの設定

1.1 真の値の設定

```
clear; clc; clf;
tic;
rng(2023);
% set true latent dimensions
kT = 3; % the number of true dimension
```

1.2 初期値の設定

まずは L を変化させるた眼，配列に確保

```
L = 10; % numLatVal = 8; % the number of latent variable dimension
k = 3;  % set true latent variable dimension
n = 150;
```

できればここで初期値の推定用アルゴリズムを入れたい．→別プログラムで．

ただし，実験の条件が変わってしまうので注意．

今回は潜在次元数を変化させて，真の値をきちんと捉えられているかを確認する．

2. 潜在構造の推定

真の値と設定した値を別に入力としないといけないので，プログラムを修正する必要あり．

```
m = 1;
d = [5, 10, 15, 20, 25, 30];
for i = d
    [X, Y, trueParams(m)] = setTrueParams(n, i, kT);
    params = setParamsDesc(trueParams(m), k, L);
    [estParams(m), history] = calcdescmetric_ver4(X, Y, k, params,
trueParams(m));
    filename = "30-resultD/num_of_D" + num2str(i) + ".mat";
    main_save_func_parfor(filename, history);
%    save(filename, "history");
    m = m + 1;
end
```

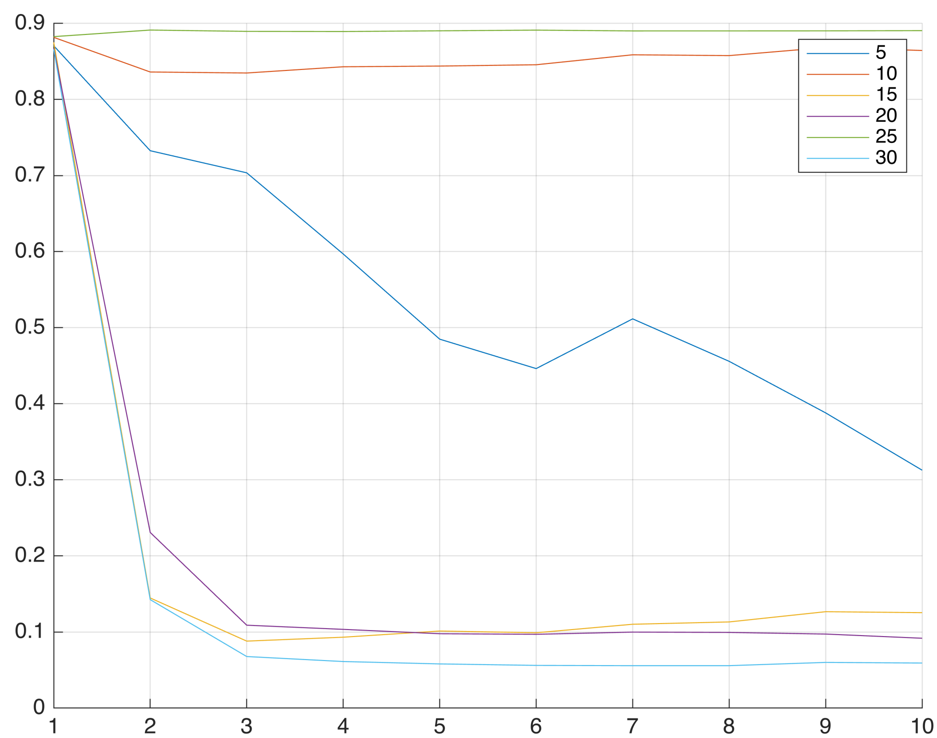
```
1 times finished    -9.734674e+03
2 times finished    -7.825862e+03
3 times finished    -7.739801e+03
4 times finished    -7.476304e+03
5 times finished    -6.846934e+03
6 times finished    -6.858855e+03
7 times finished    -6.942150e+03
```

```
8 times finished    -6.857724e+03
9 times finished    -6.571231e+03
10 times finished    -6.296592e+03
1 times finished    -9.280704e+03
2 times finished    -8.737502e+03
3 times finished    -8.632625e+03
4 times finished    -8.684588e+03
5 times finished    -8.664393e+03
6 times finished    -8.640161e+03
7 times finished    -8.631577e+03
8 times finished    -8.621938e+03
9 times finished    -8.630328e+03
10 times finished    -8.612382e+03
1 times finished    -1.141292e+04
2 times finished    -6.417534e+03
3 times finished    -5.472596e+03
4 times finished    -5.851780e+03
5 times finished    -6.091957e+03
6 times finished    -6.225169e+03
7 times finished    -6.344347e+03
8 times finished    -6.436567e+03
9 times finished    -6.506352e+03
10 times finished    -6.456461e+03
1 times finished    -1.252681e+04
2 times finished    -9.245402e+03
3 times finished    -8.296760e+03
4 times finished    -8.403853e+03
5 times finished    -8.350524e+03
6 times finished    -8.323737e+03
7 times finished    -8.388137e+03
8 times finished    -8.421968e+03
9 times finished    -8.448012e+03
10 times finished    -8.414345e+03
1 times finished    -1.306943e+04
2 times finished    -1.292922e+04
3 times finished    -1.293890e+04
4 times finished    -1.295127e+04
5 times finished    -1.294757e+04
6 times finished    -1.297209e+04
7 times finished    -1.296965e+04
8 times finished    -1.295461e+04
9 times finished    -1.294711e+04
10 times finished    -1.296555e+04
1 times finished    -1.467942e+04
2 times finished    -8.300411e+03
3 times finished    -6.869292e+03
4 times finished    -7.024809e+03
5 times finished    -7.092370e+03
6 times finished    -7.107509e+03
7 times finished    -7.131634e+03
8 times finished    -7.131458e+03
9 times finished    -7.251849e+03
10 times finished    -7.171349e+03
```

```
% 繰り返し毎にRMSEを計算するための方法に修正する必要あり！
for i = 1 : size(d, 2)
    result(i) = calcRmseDesc(trueParams(i), estParams(i));
end
```

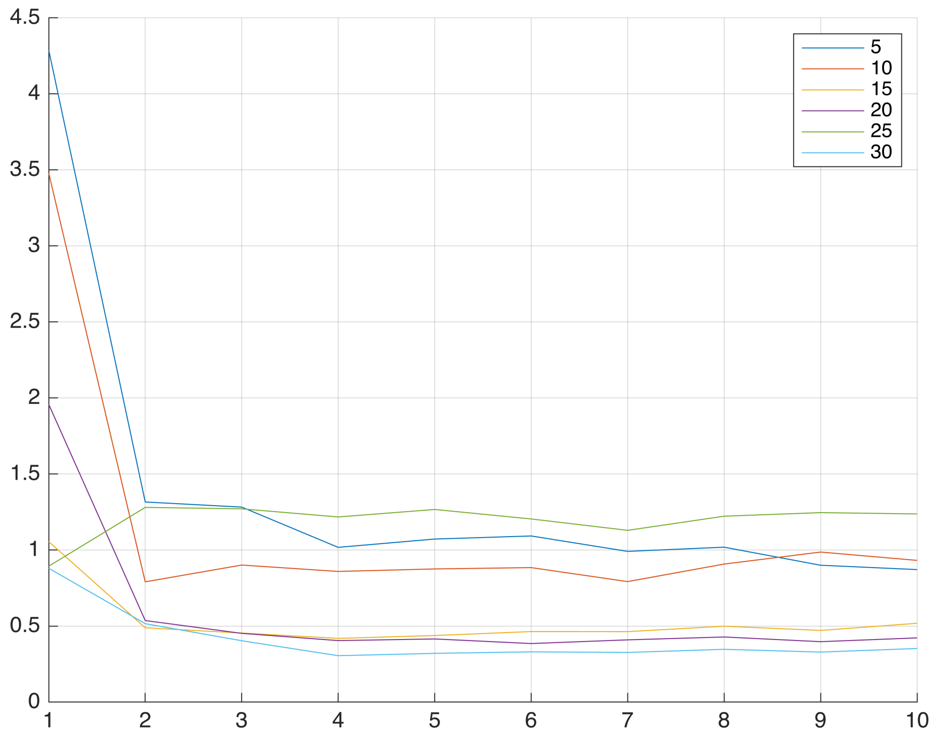
```
% ファイルの読み込み
```

```
for i = d
    str = ['history', num2str(i), '=load("30-resultD/num_of_D', num2str(i),
'.mat");'];
    eval(str);
end
% sigma の描画
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.sigma])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```

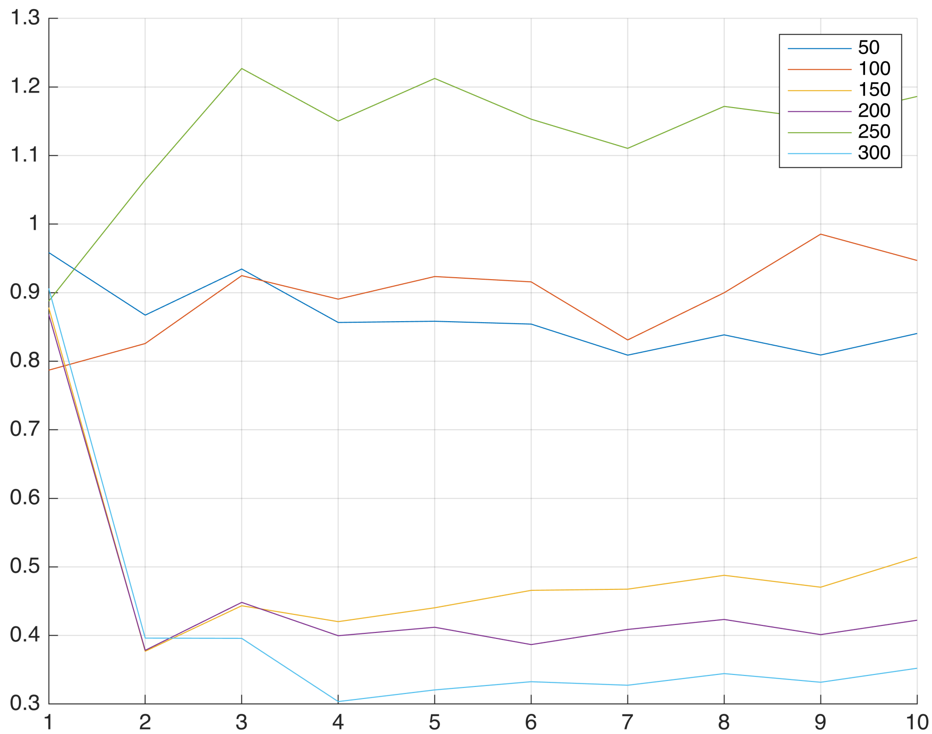



```
% Z の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.Z])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
```

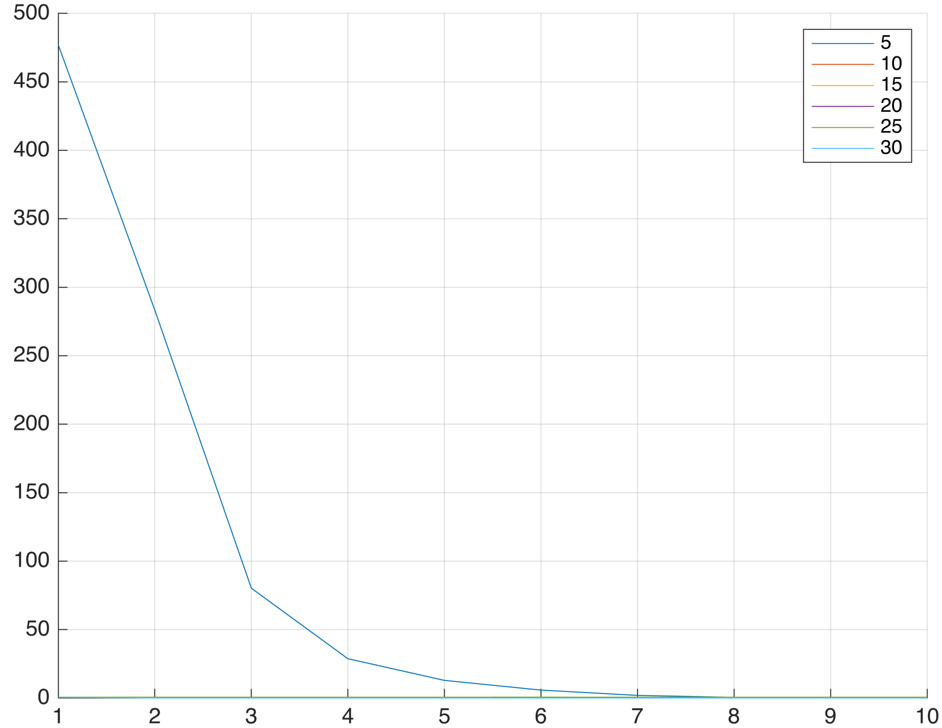
```
hold off;
```



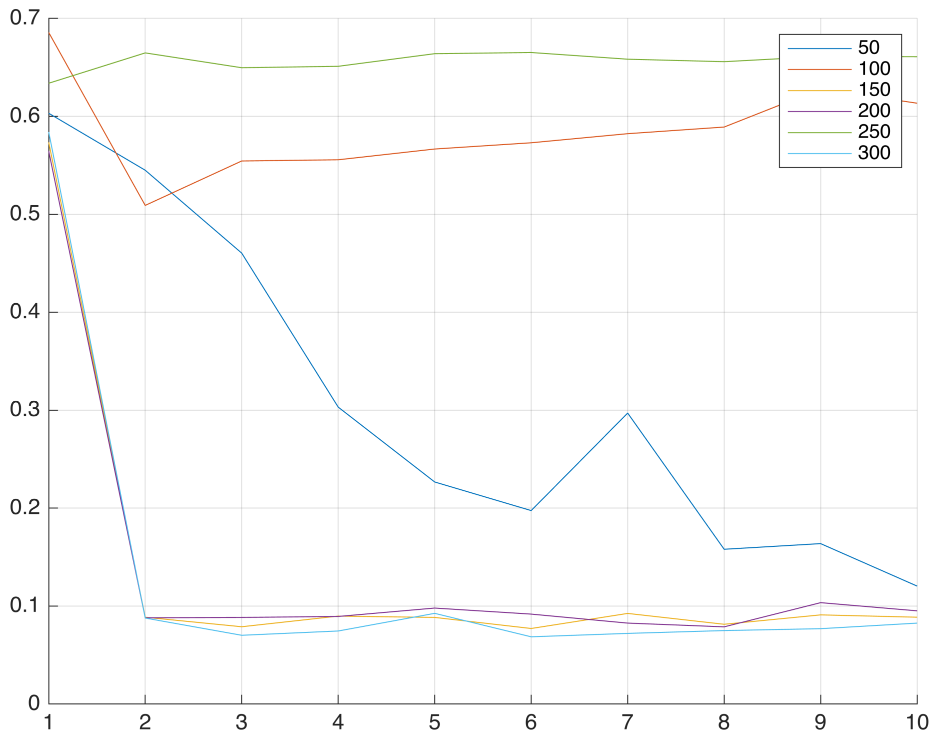

```
% Z2 の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.Z2])'];
    eval(str);
    grid on;
end
legend('50', '100', '150', '200', '250', '300');
hold off;
```



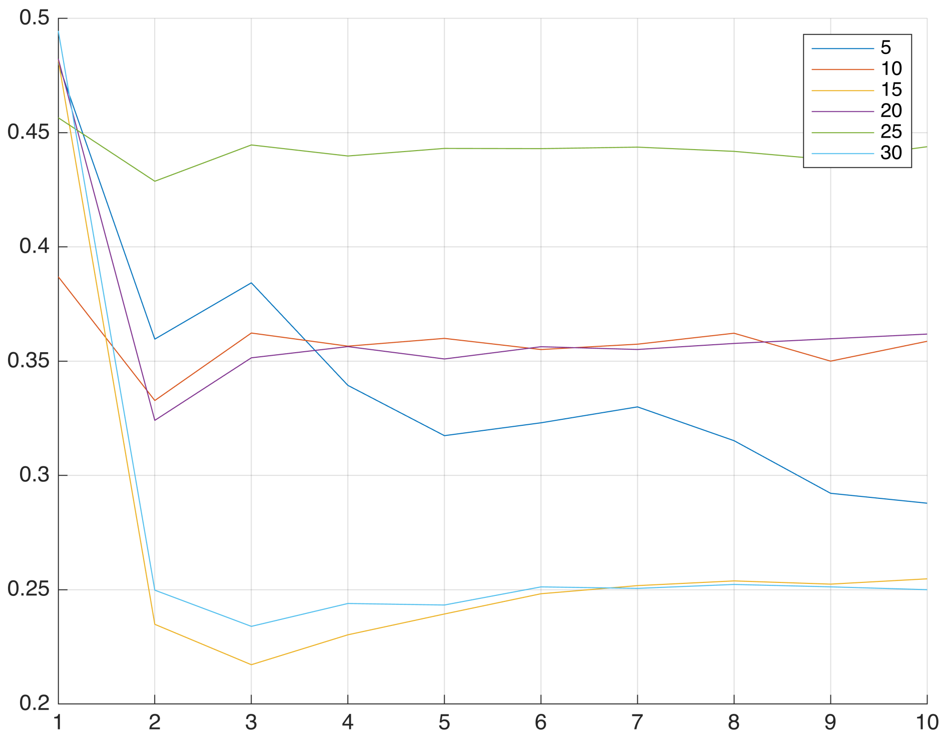

```
% F の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.F])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```



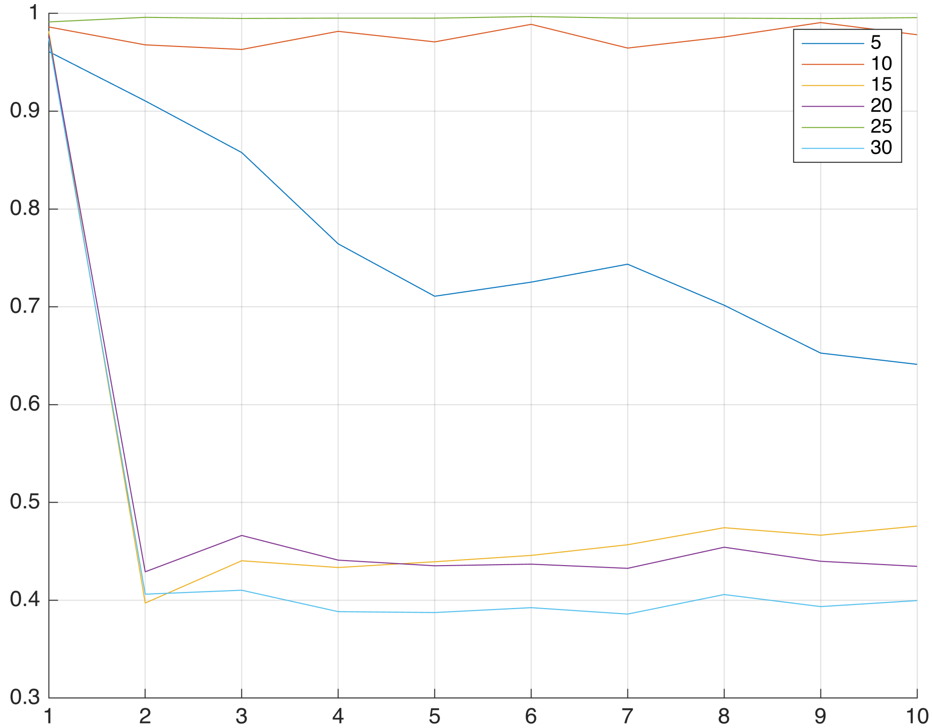

```
% F2の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.F2])'];
    eval(str);
    grid on;
end
legend('50', '100', '150', '200', '250', '300');
hold off;
```



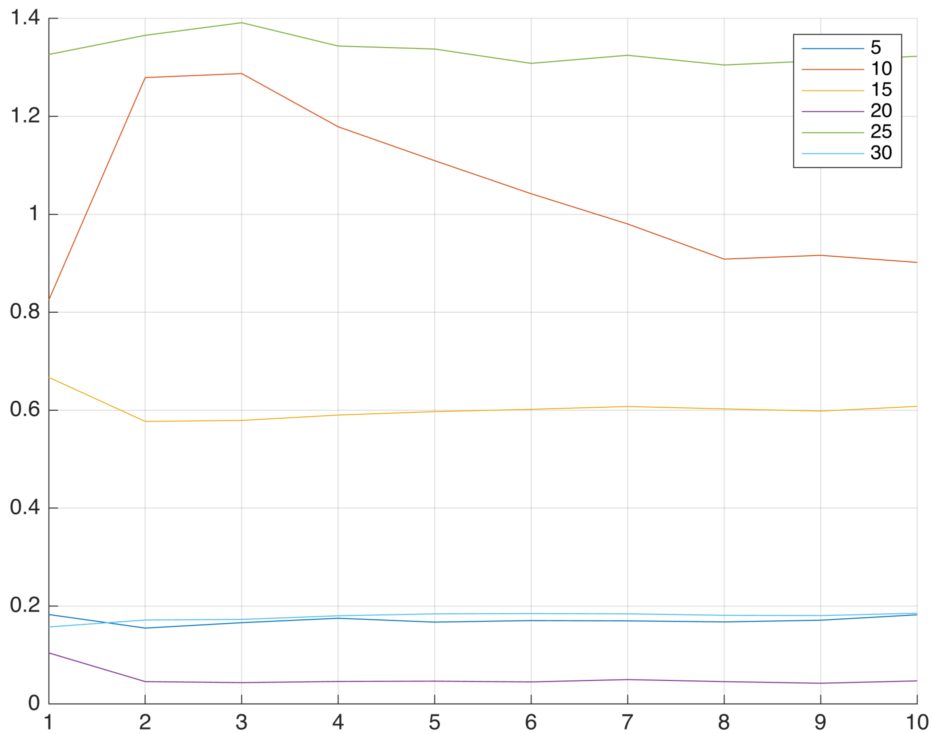

```
% Y の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.Y])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```



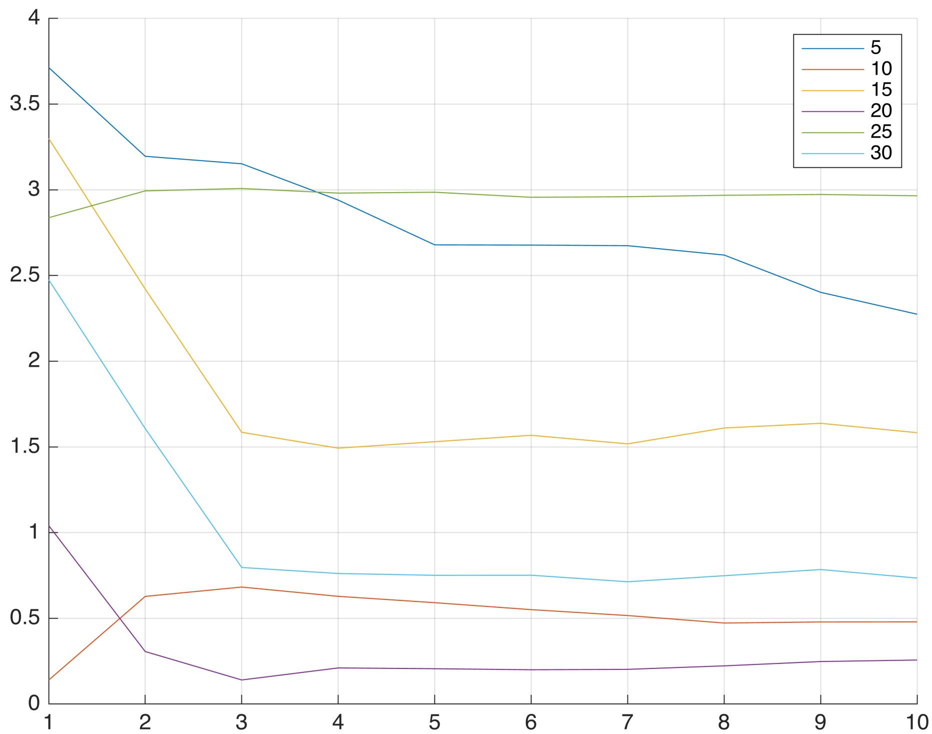

```
% X の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.X])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```



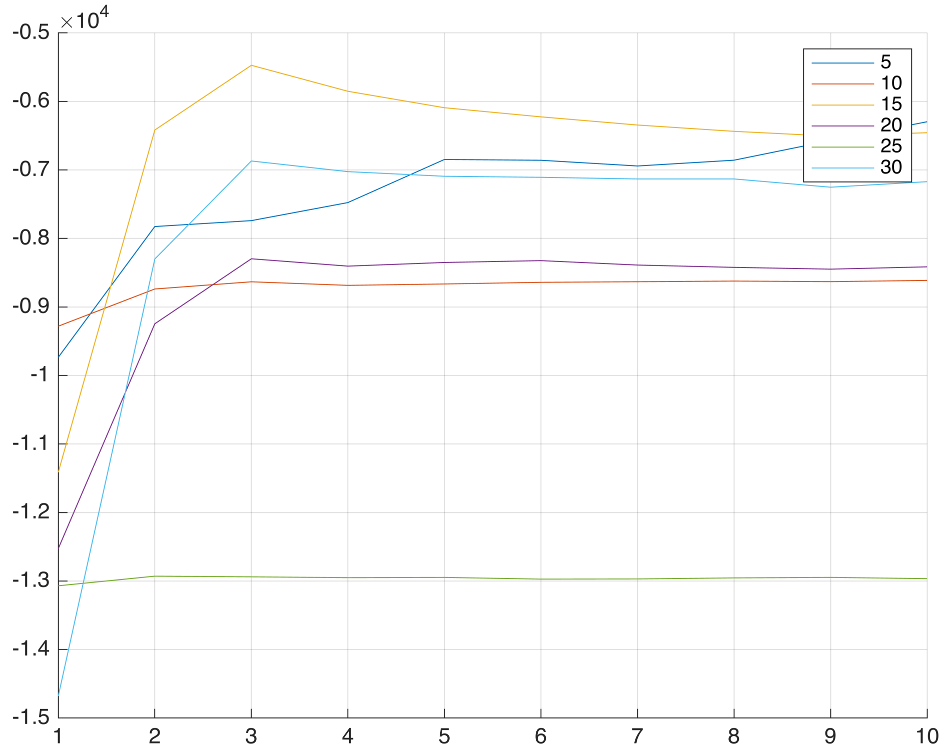

```
% w0 の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.w0])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```




```
% wの描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.w])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```




```
% Q関数の描画
clf;
hold on;
for i = d
    str = ['plot([history', num2str(i), '.history.Q])'];
    eval(str);
    grid on;
end
legend('5', '10', '15', '20', '25', '30');
hold off;
```




```
toc;
```

経過時間は 50.497842 秒です。